

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
words = ["I", "love", "deep", "learning"]
words
```

```
['I', 'love', 'deep', 'learning']
```

```
attention = np.array([
    [0.10, 0.20, 0.30, 0.40],
    [0.20, 0.30, 0.30, 0.20],
    [0.30, 0.30, 0.20, 0.20],
    [0.25, 0.25, 0.25, 0.25]
])
```

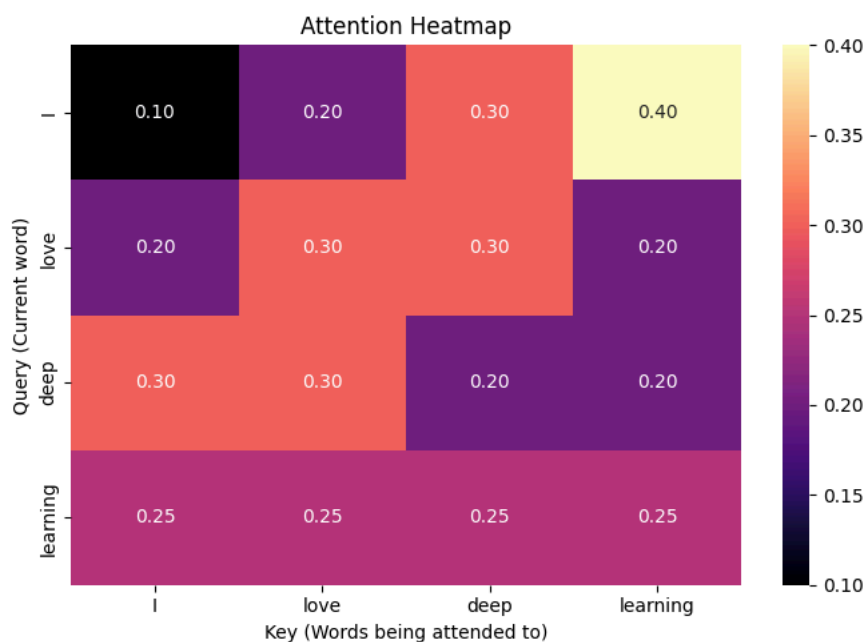
```
attention
```

```
array([[0.1 , 0.2 , 0.3 , 0.4 ],
       [0.2 , 0.3 , 0.3 , 0.2 ],
       [0.3 , 0.3 , 0.2 , 0.2 ],
       [0.25, 0.25, 0.25, 0.25]])
```

```
attention.sum(axis=1)
```

```
array([1., 1., 1., 1.])
```

```
plt.figure(figsize=(7, 5))
sns.heatmap(
    attention,
    xticklabels=words,
    yticklabels=words,
    annot=True,
    fmt='.2f',
    cmap='magma'
)
plt.title('Attention Heatmap')
plt.xlabel('Key (Words being attended to)')
plt.ylabel('Query (Current word)')
plt.tight_layout()
plt.show()
```



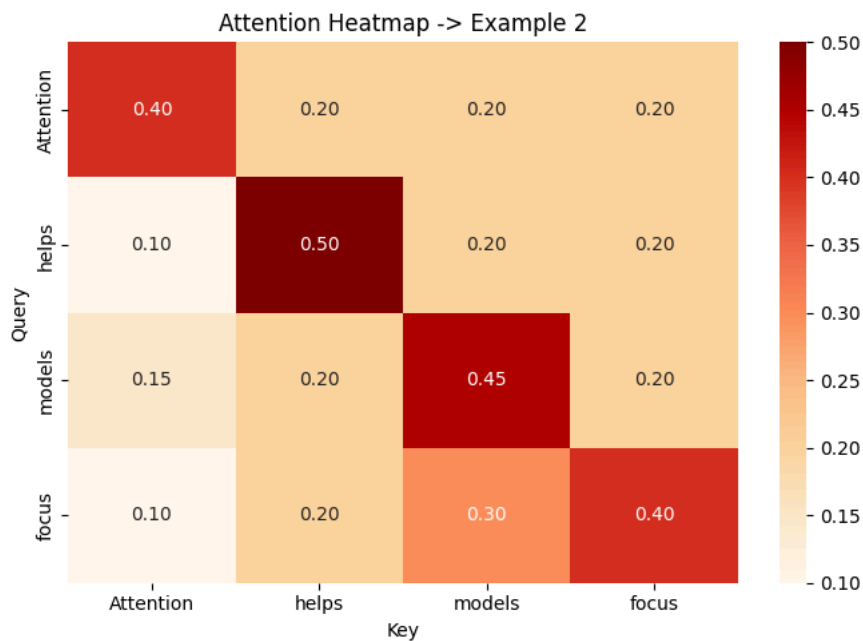
```
for i, query_word in enumerate(words):
    max_index = np.argmax(attention[i])
    print(f'Query Word : "{query_word}" attends most to "{words[max_index]}" with score {attention[i, max_index]}:
```

```
Query Word : "I" attends most to "learning" with score 0.40
Query Word : "love" attends most to "love" with score 0.30
Query Word : "deep" attends most to "I" with score 0.30
Query Word : "learning" attends most to "I" with score 0.25
```

```
words2 = ["Attention", "helps", "models", "focus"]
```

```
attention2 = np.array([
    [0.40, 0.20, 0.20, 0.20],
    [0.10, 0.50, 0.20, 0.20],
    [0.15, 0.20, 0.45, 0.20],
    [0.10, 0.20, 0.30, 0.40]
])
```

```
plt.figure(figsize=(7, 5))
sns.heatmap(
    attention2,
    xticklabels=words2,
    yticklabels=words2,
    annot=True,
    fmt='.2f',
    cmap='OrRd'
)
plt.title('Attention Heatmap -> Example 2')
plt.xlabel('Key')
plt.ylabel('Query')
plt.tight_layout()
plt.show()
```



```
my_example = ["Think", "less", "live", "more"]
```

```
my_example_attention = np.array([
    [0.25, 0.45, 0.25, 0.15],
    [0.20, 0.50, 0.40, 0.25],
    [0.15, 0.30, 0.45, 0.20],
    [0.15, 0.30, 0.10, 0.40]
])
```

```
plt.figure(figsize=(7, 5))
sns.heatmap(
    my_example_attention,
    xticklabels=my_example,
```

```
yticklabels=my_example,\nannot=True,\nfmt='.2f',\ncmap='coolwarm'\n)\nplt.title('Practice Heatmap')\nplt.xlabel('Key')\nplt.ylabel('Query')\nplt.tight_layout()\nplt.show()
```

